

Appendix: Technical Implementation

Appendix A. Reproducibility

- Repository package name: CGS410--intervener-complexity-asif
- Random seed: 13
- Random baseline replicates: 3
- Sentence cap in the current reported run: 250

Appendix B. Commands Used

```
python3 scripts/run_analysis.py --download --max-sentences 250 --random-runs 3 --seed 13
python3 scripts/run_ml_analysis.py
python3 scripts/run_llm_comparison.py --generate-prompts-only
python3 scripts/generate_course_submission.py
python3 scripts/create_github_repo_export.py
```

Appendix C. Method-to-Code Map

Report method	Main file	Purpose
Treebank loading	src/intervener_project/treebanks.py	Resolves configured treebanks
CoNLL-U parsing	src/intervener_project/conllu_io.py	Reads .conllu sentences
Intervener extraction	src/intervener_project/features.py	Builds structures and extracts interveners
Baselines	src/intervener_project/baselines.py	Creates random and simplified orders
Statistics	src/intervener_project/analysis.py	Builds summaries, Z-scores, and tests
Visualization	src/intervener_project/visualize.py	Generates report figures
Machine learning	src/intervener_project/ml.py	Trains and evaluates classifiers
Main run script	scripts/run_analysis.py	Runs the multilingual experiment
ML run script	scripts/run_ml_analysis.py	Runs logistic regression and random forest
LLM scaffold	scripts/run_llm_comparison.py	Prepares prompts and future comparison

Appendix D. Core Equations

```
I(h,d) = { i | min(pos(h), pos(d)) < pos(i) < max(pos(h), pos(d)) }
gap(h,d) = |I(h,d)|
arity(i) = number of dependents of i
subtree_size(i) = number of nodes dominated by i
complexity(i) = arity(i) + subtree_size(i)
Z = (X_real - mean(X_random)) / sd(X_random)
label(i) = 1 if complexity(i) > median(complexity), else 0
```

Appendix E. GitHub Upload Notes

- The accompanying repository folder includes code, notebooks, config, figures, tables, and the final report.
- Large raw treebanks and the 436 MB interveners.csv file are intentionally excluded from the uploadable repo.
- After uploading, replace this placeholder in the report: Public code repository:
- Current reported dataset size: 31,007 real dependencies and 44,881 real interveners.